Course Overview

Python for AI-Driven Automation and Business Data Science

Course materials

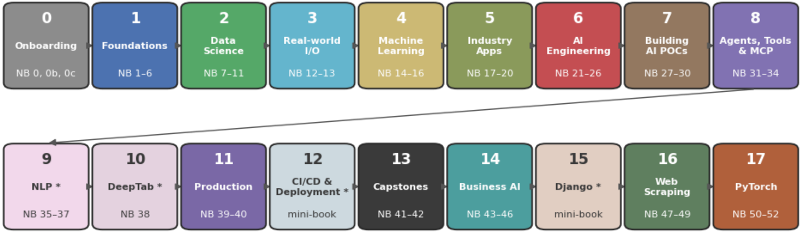
Contents

Outline

The course in one diagram

Python for AI-Driven Automation and Business Data Science

52 lessons + 5 labs + 13 appendices · modules 0–17 · self-paced — spiral order

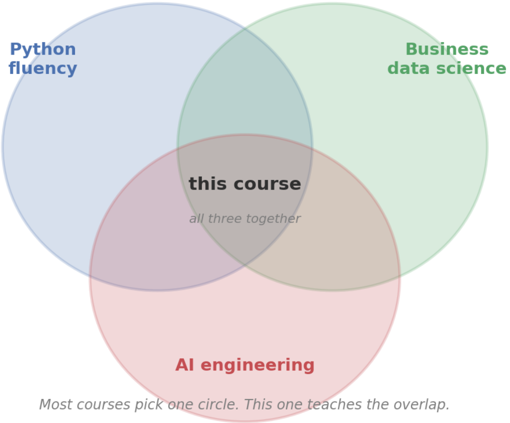


* optional — NLP, DeepTab, CI/CD and Django can be skipped or read as reference

— runs entirely offline · every exercise has a worked solution · every notebook executes end-to-end —

Foundations of every page: read prose → run code → tweak → predict → re-run

Where this course sits



Most courses pick one circle. This one teaches the overlap.

Three skills, taught together

1. Python fluency

Read, write, and modify clean Python without friction. Functions, modules, error handling, defensive parsing.

2. Working with data professionally

pandas, NumPy, matplotlib, scikit-learn, statistics, time-series forecasting — plus PyTorch deep learning for when the classical toolkit stops. Real datasets, not toy ones.

3. AI-driven automation

Calling LLMs as functions. Structured output. Retrieval-augmented generation. Tools and agents. Document processing. *Plus* the engineering wrapper around all of it.

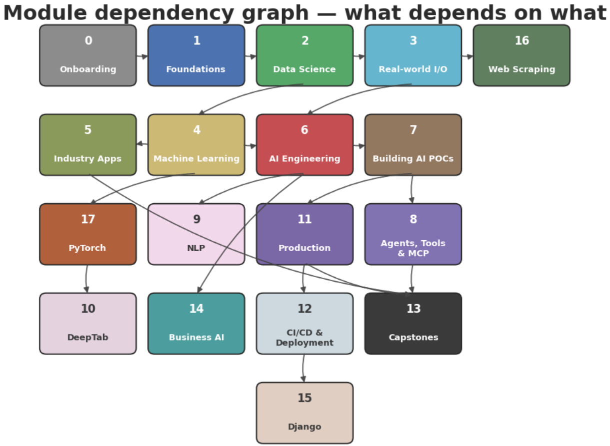
Outline

Eighteen modules — 52 lessons, 5 labs (+ 13 appendices)

#	Module	Notebooks
0	Onboarding	master onboarding + overview + see-it-work
1	Foundations	NB 1–6 — Python you can read without friction
2	Data science	NB 7–11 — pandas, NumPy, plots, stats, time series
3	Real-world I/O	NB 12–13 — HTTP, SQL, validation
4	Machine learning	NB 14–16 — sklearn, model evaluation, features
5	Industry applications	NB 17–20 — churn value, fraud, segmentation, demand
6	AI engineering	NB 21–26 — LLM patterns, RAG, agents, evaluation
7	Building AI POCs	NB 27–30 — setup, three POCs, RAG deep dive, agentic AI
8	Agents, tools & MCP	NB 31–34 — architectures, robust tools, MCP, multi-agent
9	NLP (optional)	NB 35–37 — topic modeling, sentiment
10	DeepTab (optional)	NB 38 — deep learning for tabular data
11	Production	NB 39–40 — packaging, scheduling, config & secrets
12	CI/CD & deployment	mini-book + 3 labs — Docker, GitHub Actions, HTTPS
13	Capstones	NB 41–42 — analytics + AI assistant
14	Business AI	NB 43–46 — strategy, architecture, BPM, governance
15	Django (optional)	mini-book + 2 labs — serve a model behind a web UI
16	Web scraping	NB 47–49 — BeautifulSoup, Firecrawl, the OpenAlex API
17	Deep learning (PyTorch)	NB 50–52 — tensors & the training loop, training craft, embeddings & serving

Every module folder has its own README — goals, prerequisites, what you'll build. Read it before the module's first notebook. Module 17 slots pedagogically right after Module 4.

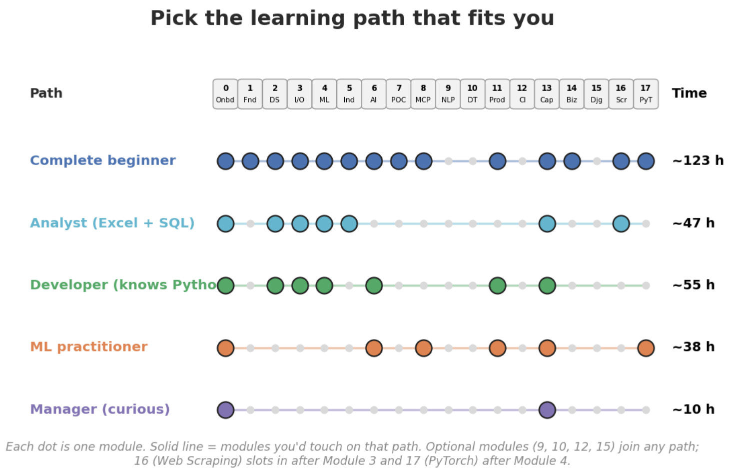
Module dependencies — what comes before what



Edges flow downstream. Module N is safe to start once every upstream module is comfortable.

Outline

Pick the path that fits you



When to pick which path

Complete beginner

Never written code before. Take all modules in **spiral order** (see it work first, then the skills, then the builds). Time-budget: **about 123 hours** — the interactive estimator in 00b computes yours.

Analyst who knows Excel and SQL

Skim Module 1 (Python basics); Module 3's SQL will feel familiar. Focus on Modules 2 (data science), 4 (ML), 5 (industry applications), 16 (web scraping), 13 (capstone). Time-budget: **about 47 hours**.

ML practitioner adding AI engineering

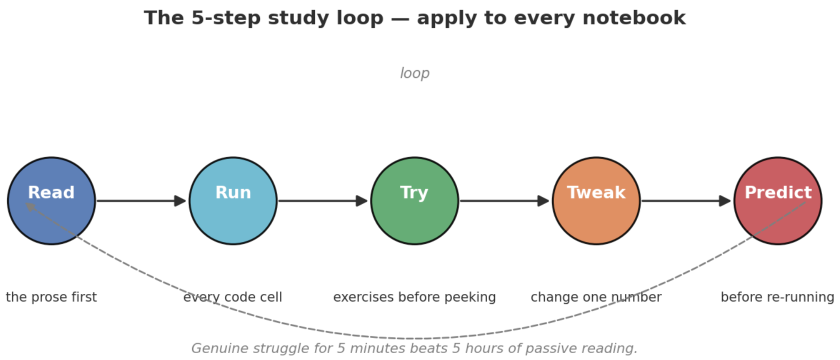
Skip foundations and data science. Focus on Modules 6 (AI engineering), 8 (agents & MCP), 17 (PyTorch), 11 (production), 13 (capstone). Time-budget: **about 38 hours**.

Twelve weeks, self-paced

Got more time? Move faster. Got less? Split one week into two.

Outline

The five-step study loop



Apply this loop to *every* notebook. Genuine struggle beats passive reading.

Eight visual markers you'll see throughout

Marker	Meaning
`(Tip)`	A small "stop and notice" callout.
`(Intuition)`	The mental model behind the syntax.
`(Pitfall)`	A bug or anti-pattern that bites if you're not warned.
`(Quick exercise)`	A ∼2-minute in-lesson checkpoint with a collapsible solution.
`(Exercises)`	Hands-on tasks. Try *before* peeking at the solution.
`(Bonus)`	One larger applied task per notebook.
`(Debug me)`	A cell that *intentionally* contains a bug for you to find.
`(Next step)`	Always points to the next notebook in your path.

Once you internalise the markers they become road signs, not decoration.

Checkpoints, quizzes, and the fast track

317 in-lesson checkpoints — the rhythm of every lesson

Roughly every ∼20 minutes of material: a `(Quick exercise)` with a scaffolded starter and a collapsible solution — every solution executed in a fresh kernel. Teach → try for ∼2 minutes → reveal.

15 module quizzes

Five multiple-choice questions per content module (∼10 minutes) in `quizzes/` — read code, predict behaviour, pick the right tool. Score 0–2/5? Re-do the module before moving on.

Short on time? The fast track

`fast_track/` condenses the whole course into 22 trimmed lessons (∼26 hours) — the 14 core essentials plus 8 breadth extensions.

Outline

Six small artefacts you ship along the way

Six small artefacts you'll ship by the end

KPI snapshot
An AI-bot KPI report in ~30 lines of Python

Live API ETL
Daily weather pulled from Open-Meteo

SQL report
Channel summary written in SQL + pandas

Forecast
3-month forecast with Holt-Winters

Inbox triage
LLM that classifies + summarises feedback

Scheduled job
Weekly report task with retries + alerts

Plus two big capstones at the end

Capstone A — AI Support-Bot Analytics (NB 41)

Five channels $\times$ twelve months of operational data $\rightarrow$ 2$\times$2 executive dashboard $\rightarrow$ regression demonstrating Simpson's paradox $\rightarrow$ five-bullet executive summary.

Capstone B — AI Customer-Feedback Assistant (NB 42)

Ingest free-form customer feedback $\rightarrow$ classify $+$ extract via the MockLLM $\rightarrow$ ground answers in a knowledge base via RAG $\rightarrow$ orchestrate as a scheduled job with alerts. End-to-end AI feature.

Outline

LLM providers — local and hosted, four options

Provider	Class	Best for
OpenAI	`OpenAILLM("gpt-5.4-mini")`	Reliable default
Anthropic	`AnthropicLLM("claude-haiku-4-5")`	Long context, careful tone
Google	`GoogleLLM("gemini-2.5-flash")`	Cheapest at high volume
Ollama	`OllamaLLM("llama3.2:3b")`	**Local** – no internet, no key

Default = the offline MockLLM

Every AI notebook runs offline by default. Swapping in a real provider is a *one-line* change thanks to the unified interface in `llm_providers.py` at the repo root.

See the dedicated guide

Notebook `06_ai_engineering/A1_llm_providers_guide.ipynb` has the full setup walk-through, model recommendations, and cost estimates.

What you need to install

For Google Colab

Nothing. All required libraries are pre-installed. Open the notebook, hit Shift+Enter.

For local Jupyter

```
python -m venv .venv
source .venv/bin/activate
pip install -r requirements.txt
jupyter lab
```

Notebook 0 includes an environment-check cell you run to confirm everything works before you move on.

Now → open Notebook 0

One file to open. The rest unfolds from there.

`00_onboarding/00_master_onboarding.ipynb`

Three things that notebook does for you

1. Confirms your Python environment is correctly set up.
2. Explains the five-step study loop you'll use throughout.
3. Helps you pick the right learning path from the five options.

One last reminder before you start

Five-step study loop, in your pocket

Read the prose first → **Run** every code cell → **Try** the exercises before peeking → **Tweak** one number → **Predict** the new output before re-running.

When to use which marker

`(Tip)` information; `(Intuition)` mental model; `(Pitfall)` avoid this; `(Quick exercise)` ~2-minute checkpoint; `(Exercise)` hands-on practice; `(Bonus)` bigger applied task; `(Debug me)` intentional bug — find it.

When you're stuck for 15 minutes

Copy the full error into a search engine. 95% of beginner errors are well-documented. The course is designed to teach you to debug, not to spare you the experience.

Welcome.

111 notebooks await you.

Open Module 0 next.